

Comprehensive Curricular Analysis and Deterministic Blueprint of Artificial Intelligence Examination Patterns

The rigorous evaluation of academic assessments within the domain of Artificial Intelligence (AI) provides critical visibility into the pedagogical priorities and cognitive expectations placed upon emerging computer science engineers. By systematically analyzing the examination frameworks from recent academic cycles—specifically the June/July 2024, December 2024/January 2025, June/July 2025, and December 2025/January 2026 sessions—a distinct topology of core competencies emerges. This analysis targets the structured examination schema designated as BAD402 under the Choice Based Credit System (CBCS) framework.¹ The objective is to unearth the frequency, thematic distribution, and cognitive depth of assessed topics to provide an exhaustive inventory of historical inquiries and a deterministic blueprint of topics anticipated for future assessments.

The curriculum dictates a profound cognitive transition. It demands that candidates evolve from understanding deterministic, symbolic logic paradigms to mastering stochastic, probabilistic reasoning frameworks.³ The subsequent sections provide an exhaustive, module-by-module deconstruction of every individual question posed across these four assessment cycles. By mapping their historical frequency and the specific temporal instances of their occurrence, this report isolates the mechanical, algorithmic, and theoretical expectations of the examiners. Following this granular taxonomy, a synthesizing framework delineates the latent academic trajectories, ultimately yielding a comprehensive matrix of topics upon which all future examinations will be formulated.

Module 1: Introduction to Artificial Intelligence and Intelligent Agents

The initial module establishes the philosophical, historical, and structural foundations of artificial intelligence. It transitions the academic focus from defining the abstract parameters of machine intelligence to the structural engineering of intelligent agents interacting within specific computational and physical environments.³ The pedagogical intent is to certify that candidates understand not merely what AI is, but how intelligent systems are architected to perceive and act rationally.

Exhaustive Inventory of Historical Inquiries (Module 1)

The following table categorizes every discrete question presented in the BAD402 examinations for Module 1, detailing the specific inquiry, its temporal occurrence, historical frequency, and cognitive classification.¹

Exact or	Temporal	Frequency	Bloom's Level /
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Paraphrased Examination Question	Occurrence		Course Outcome
Define Artificial Intelligence. Explain the foundation of AI in detail.	Jan 2026 (Q1a), July 2024 (Q1a)	2	L1, L2 / CO1
Explain the historical development of AI, highlighting key milestones and breakthroughs.	Jan 2026 (Q1b)	1	L2 / CO1
Explain all four different approaches to AI in detail (acting humanly, thinking humanly, thinking rationally, acting rationally).	July 2024 (Q1b)	1	L1 / CO1
What is AI? List out the applications of AI, state the characteristics of AI problems.	July 2025 (Q2a)	1	L1 / CO1
Compare and contrast human intelligence to artificial intelligence with numerous examples and applications.	July 2025 (Q1b)	1	L4 / CO1
Explain the	Jan 2025 (Q1a)	1	L2 / CO1

significance of the Turing Test in AI. What abilities does a computer need to pass the Turing test? Discuss why AI researchers have not focused extensively on passing it.			
Give PEAS specification for: i) Automated taxi driver ii) Medical diagnostic system.	July 2024 (Q2a)	1	L1 / CO1
Analyze and discuss PEAS descriptor for the following applications in detail: i) Medical diagnosis system ii) Taxi driver iii) Interactive English tutor iv) Part picking robot v) Refinery controller.	Jan 2025 (Q3b)	1	L3 / CO1
Explain the following: i) PEAS.	July 2025 (Q1c)	1	L2 / CO1
Briefly explain the properties of Task Environment.	Jan 2026 (Q2a)	1	L2 / CO1
Differentiation: i) Fully observable vs partially observable ii) Single agent vs Multiagent iii)	July 2024 (Q2b)	1	L1 / CO1

Deterministic vs stochastic iv) Static vs Dynamic.			
Write the percept sequence for a vacuum cleaner agent and tabulate the workflow of the same with respect to the scenario with location of square A and B.	Jan 2025 (Q2a)	1	L3 / CO1
Compare simple reflex agents and model-based reflex agents, focusing on their perception processing, decision-making methods, and explain how model-based agents address limitations.	Jan 2025 (Q2a OR)	1	L3 / CO1
Explain the following with respect to structure agents: (i) Simple reflex (ii) Model based reflex (iii) Utility based.	Jan 2026 (Q2b)	1	L2 / CO1
Explain the following: ii) Simple reflex agent iii) Model based agent.	July 2025 (Q1c)	1	L2 / CO1

Explain the structure of agents and analyze the characteristics of intelligent agents.	July 2025 (Q2c)	1	L2 / CO1
Analyze and generalize what is a rational agent.	July 2025 (Q2b)	1	L4 / CO1

Theoretical Analysis and Evaluative Trends

The empirical data demonstrates a pronounced emphasis on the conceptual architecture of intelligent systems. The epistemological foundations of AI are frequently tested through definitions and historical milestones.¹ Candidates are expected to articulate the four primary approaches to AI: systems that think like humans, systems that act like humans, systems that think rationally, and systems that act rationally.¹ The inclusion of the Turing Test in the January 2025 paper further solidifies this expectation, demanding that students enumerate the specific sub-disciplines required to pass the test, namely Natural Language Processing, Knowledge Representation, Automated Reasoning, Machine Learning, Computer Vision, and Robotics.¹ Beyond philosophy, the performance evaluation frameworks are heavily scrutinized. The Performance, Environment, Actuators, and Sensors (PEAS) framework is a highly recurring metric, indicating that evaluators prioritize the practical mapping of theoretical agents to real-world scenarios.¹ Assessors consistently provide specific hypothetical systems—ranging from automated taxis to refinery controllers—and require the candidate to generate a highly specific PEAS matrix.

Simultaneously, the architectural distinctions between agent topologies form a core comparative analysis that appears in almost every iteration.¹ Candidates must possess the technical vocabulary to differentiate a simple reflex agent, which operates solely on condition-action rules based on the current percept, from a model-based reflex agent, which maintains an internal state to track unobservable aspects of the environment.¹ Advanced inquiries extend this to utility-based agents, which integrate a performance measure to maximize operational happiness or efficiency.¹ The cognitive demand within this module is broad, beginning with foundational recall (Bloom's Level L1/L2) and escalating to high-order analytical tasks, such as contrasting human versus artificial intelligence at Bloom's Level L4.¹

Module 2: Problem-Solving and Uninformed Search Strategies

Module 2 represents the transition from abstract agent definitions to the computational mechanics of state-space exploration. The core pedagogical objective is to evaluate a

candidate's proficiency in translating real-world ambiguities into mathematically rigorous problem formulations, and subsequently deploying uninformed (blind) search algorithms to navigate these complex state spaces.³

Exhaustive Inventory of Historical Inquiries (Module 2)

The structural integrity of a student's algorithmic comprehension is tested via the following inquiries, which form the entire historical corpus for Module 2.¹

Exact or Paraphrased Examination Question	Temporal Occurrence	Frequency	Bloom's Level / Course Outcome
Explain five components of a well-defined problem. Consider an 8-puzzle problem as an example and explain.	July 2024 (Q3a)	1	L2 / CO2
What are the four components to define a problem? Define them.	July 2025 (Q1a)	1	L1 / CO1
Explain Goal Formulation and Problem Formulation with examples.	Jan 2026 (Q4b)	1	L2 / CO2
Explain toy problems and real-world problems in the context of problem-solving approaches with an example for each	Jan 2025 (Q4a)	1	L2 / CO2

type in detail.			
Compare and contrast the vacuum world problem and the 8-tile puzzle problems discussing their state representations, initial states, actions and goal tests.	Jan 2025 (Q4b)	1	L3 / CO2
You are given two jugs, a 5 liters one and a 4 liters one... How can you get exactly 2 liters of water in the 5 liters jug? Apply water Jug problem algorithm.	July 2025 (Q3a)	1	L3 / CO2
Discuss how problem-solving agents interact with their environments.	Jan 2026 (Q3a)	1	L2 / CO2
Explain the components and architecture of a problem-solving agent.	Jan 2025 (Q5a)	1	L2 / CO2
Discuss in detail the Infrastructure for search algorithms.	July 2024 (Q3b)	1	L2 / CO2
Discuss the	Jan 2026 (Q4a)	1	L2 / CO2

different solutions and metrics for searching.			
Compare and contrast depth-first search with breadth-first search with examples.	Jan 2025 (Q5b)	1	L3 / CO2
Explain the principles of breadth-first search as a problem solving strategy with an example.	Jan 2026 (Q3b)	1	L2 / CO2
Write an algorithm for Breadth-first search and explain with an example.	July 2024 (Q4a)	1	L2 / CO2
Explain Breadth First Search (BFS) algorithm and apply BFS to find the solution for the above graph. Also find the optimum path and cost.	July 2025 (Q4a)	1	L3 / CO2
Explain Depth first search techniques in detail.	July 2024 (Q4b)	1	L2 / CO2
Describe Depth First Search (DFS) search algorithm	July 2025 (Q3b)	1	L2 / CO2

with an example.			
Describe the iterative deepening depth first search with an example.	July 2025 (Q4b)	1	L2 / CO2

Theoretical Analysis and Evaluative Trends

Uninformed search paradigms are the undisputed focal point of the second module. Breadth-First Search (BFS) and Depth-First Search (DFS) are universally tested across the assessment cycles, signaling their status as non-negotiable computational competencies.¹ Evaluators do not merely seek a superficial definition of these algorithms; they require a deep understanding of the underlying data structures. Candidates must articulate that BFS utilizes a First-In-First-Out (FIFO) queue architecture to explore the shallowest unexpanded nodes, guaranteeing optimality in step-costs, whereas DFS utilizes a Last-In-First-Out (LIFO) stack architecture to plunge to the deepest nodes, offering memory efficiency at the expense of optimality and completeness.¹

The testing modality frequently bifurcates into two forms: theoretical algorithmic explanation and applied graph traversal. For instance, the July 2025 assessment mandated candidates to apply BFS to a specific multi-nodal graph, extracting the optimal path and calculating its associated cost natively.¹ Iterative Deepening Depth-First Search (IDDFS) was also explicitly tested, requiring students to explain how it synthesizes the memory efficiency of DFS with the algorithmic completeness of BFS by incrementally expanding the depth limit.¹

A compelling and highly prevalent trend is the persistent use of classical AI paradoxes to test state-space formulation. Assessors demand that students deconstruct abstract puzzles into the five core components of a problem: initial state, actions, transition model, goal test, and path cost.¹ The 8-puzzle, the vacuum world, and the Water Jug problem serve as foundational execution models.¹ The Water Jug problem, appearing in July 2025, specifically required candidates to calculate the exact algorithmic derivation to isolate exactly two liters of water using five-liter and four-liter ungraduated vessels.¹ This necessitates writing out state

transitions as coordinate pairs (X, Y) and applying specific production rules (e.g., filling, pouring, emptying) until the goal state is achieved.⁹ This shift from declarative knowledge to procedural execution underscores the mathematical rigor expected within the curriculum.

Module 3: Informed Search and Logical Agents

While Module 2 restricts agents to blind exploration, Module 3 introduces heuristic functions, transforming search algorithms from brute-force mechanisms into optimized, cost-aware systems. Additionally, the module pivots into symbolic AI, introducing propositional logic as a preliminary knowledge representation methodology.⁸

Exhaustive Inventory of Historical Inquiries (Module 3)

The examination schema for Module 3 tests the confluence of mathematical heuristics and foundational symbolic logic through the following specific questions.¹

Exact or Paraphrased Examination Question	Temporal Occurrence	Frequency	Bloom's Level / Course Outcome
Define informed search strategies in the context of AI. Difference between Informed and Uninformed search strategies.	Jan 2026 (Q5a)	1	L2 / CO3
Compare blind search and heuristic search algorithm in detail.	July 2025 (Q5a)	1	L4 / CO3
Explain the following concepts with example: i) Heuristic function.	July 2025 (Q6c)	1	L2 / CO3
Explain A* algorithm. Give one example where A* is suitable to apply.	Jan 2026 (Q5b)	1	L2 / CO3
Explain the A* search to minimize the total estimated cost.	July 2024 (Q5a)	1	L3 / CO3
Apply A* search algorithm to find the solution path	Jan 2025 (Q5b Mod3)	1	L3 / CO3

from the start node (S) to the goal node (G). The heuristic values (h) and travel costs (C) are provided.			
In the below graph, find the path from A to G using Greedy Best First search and A* search algorithm.	July 2024 (Q6a)	1	L3 / CO3
Describe A* search algorithm with an example.	July 2025 (Q6a)	1	L3 / CO3
Describe the principles of greedy best first search as an informed search strategy. How does it make use of heuristic information?	Jan 2026 (Q6a)	1	L2 / CO3
Write an algorithm for hill climbing search and explain in detail.	July 2024 (Q5b)	1	L3 / CO3
Solve the following eight-tile puzzle using heuristic function approach and the tree diagram considering the initial and final	Jan 2025 (Q6b)	1	L2 / CO3

states.			
Outline a generic knowledge-based agent's program and discuss the difference between declarative and procedural approaches.	Jan 2025 (Q5a Mod3)	1	L3 / CO3
Describe the Wumpus world environment and the PEAS specification for the knowledge based agent. Explain how the agent navigates.	Jan 2025 (Q6a)	1	L2 / CO3
Write a note on Wumpus world problem.	July 2025 (Q5b)	1	L2 / CO3
Explain the syntax and semantics of propositional logic.	July 2024 (Q6b)	1	L3 / CO3
Write the connectives used to form complex sentence of propositional logic. Give example for each.	July 2025 (Q5c)	1	L2 / CO3
Compare proposition logic and predicate logic	July 2025 (Q6b)	1	L4 / CO3

in detail with example.			
Explain the following with examples: (i) Logical Equivalence (ii) Inference Rules (iii) Horn Clauses.	Jan 2026 (Q6b)	1	L2 / CO3
Explain the following concepts with example: ii) Atomic sentence iii) Complex sentence.	July 2025 (Q6c)	1	L2 / CO3

Theoretical Analysis and Evaluative Trends

The algorithmic cornerstone of Module 3 is indisputably the A* search algorithm.¹ The frequency and depth of its evaluation indicate that candidates must possess absolute mastery over the mathematical function $f(n) = g(n) + h(n)$, where $g(n)$ denotes the exact

path cost from the start node and $h(n)$ denotes the estimated heuristic cost to the goal.¹⁰ The examinations consistently require candidates to differentiate A* from Greedy Best-First

Search, which operates solely on $f(n) = h(n)$ and thus abandons optimality in favor of speed.¹

Evaluators do not allow candidates to rely entirely on algorithmic definitions. The January 2025 and July 2024 examinations explicitly required the application of A* and Greedy Best-First Search to mapped networks featuring predefined heuristic arrays.¹ The candidate must

manually trace the node expansion, calculating $f(n)$ at each frontier and proving the minimal path. Furthermore, the heuristic approach is applied to state-space trees, as seen in the January 2025 requirement to solve the eight-tile puzzle using heuristic functions.¹ This requires knowledge of specific heuristics like Manhattan distance or the misplaced tiles calculation to prove algorithmic admissibility.¹⁰

Simultaneously, the theoretical wing of this module transitions heavily into logical agents. The Wumpus World paradigm acts as the canonical proving ground for knowledge-based agents acting on percepts such as stench, breeze, and glitter.¹ Assessors consistently probe the epistemological divide between propositional logic and predicate logic, demanding that

students articulate syntax, semantics, and the specific use of logical connectives.¹ The inclusion of questions addressing atomic sentences, complex sentences, logical equivalence rules, and Horn clauses indicates a rigorous prerequisite understanding required before candidates can computationally manipulate First-Order Logic in the subsequent module.¹

Module 4: First-Order Logic and Inference

Mechanisms

Module 4 elevates the constraints of propositional logic by introducing First-Order Logic (FOL), permitting the expression of complex relationships through objects, predicates, and universal or existential quantifiers. The focus subsequently moves to the automation of reasoning via forward chaining, backward chaining, and resolution methodologies.³

Exhaustive Inventory of Historical Inquiries (Module 4)

The cognitive transition from declarative definitions to rigorous mathematical logic and automated proof structures defines the Module 4 examination schema.¹

Exact or Paraphrased Examination Question	Temporal Occurrence	Frequency	Bloom's Level / Course Outcome
Explain the syntax and semantics of the first order logic.	July 2024 (Q7a)	1	L2 / CO2
Explain the propositional syntax and semantics of First Order Logic (FOL).	Jan 2026 (Q7b)	1	L2 / CO4
What are predicates? Explain its syntax and semantics.	July 2025 (Q7a)	1	L2 / CO4
Explain the following with respect to first-order logic: (i) Assertions and	Jan 2025 (Q7a), July 2024 (Q7b)	2	L2 / CO2, CO4

queries (ii) Numbers, sets and lists (iii) The wumpus world / Kinship domain.			
Define universal and existential instantiations with examples.	Jan 2025 (Q7a), July 2025 (Q7b)	2	L1, L2 / CO4
Write appropriate quantifiers for the following: Some students read well, Some students like some books, Some students like all books, All students like some books, All students like no books.	July 2025 (Q8a)	1	L3 / CO4
Apply predicate logic to translate and formalize the following statements: Marcus was a man, Marcus was a Pompeian, All Pompeian were Romans, Caesar was a ruler...	Jan 2025 (Q8a)	1	L3 / CO4
Provide examples of how inference can be applied to draw conclusions in a given knowledge base represented	Jan 2026 (Q7a)	1	L2 / CO4

FOL.			
Explain unification and lifting in detail.	July 2024 (Q8a)	1	L3 / CO4
Explain the process of unification.	July 2025 (Q8b)	1	L3 / CO4
Explain the concept of resolution in first order logic with appropriate procedure.	July 2025 (Q8a)	1	L3 / CO4
Explain backward chaining algorithm with an example.	Jan 2025 (Q8b)	1	L2 / CO4
Outline the process of backward chaining in FOL. Provide examples to illustrate how it works.	Jan 2026 (Q8b)	1	L2 / CO4
Explain Forward chaining algorithm with an example.	July 2024 (Q8b)	1	L3 / CO4
Describe the principles of forward chaining in FOL. Provide examples to illustrate how it works.	Jan 2026 (Q8a)	1	L2 / CO4
Write and explain simple backward chaining algorithm	July 2025 (Q8b)	1	L3 / CO4

and forward chaining algorithm for first order knowledge bases with example.			
Prove the following using Backward and forward chaining: "Solan is a criminal" based on a provided text about hostile nations and missile sales.	Jan 2025 (Q7b)	1	L2 / CO4
Consider the following knowledge base regarding Gita and food... Goal: Prove Gita likes almond.	July 2025 (Q7c)	1	L3 / CO4

Theoretical Analysis and Evaluative Trends

The syntax of First-Order Logic is rigorously tested through semantic translation. The classic "Marcus was a Pompeian" taxonomy of statements appeared in the January 2025 evaluation, requiring students to extract predicates, quantifiers, variables, and connectives accurately to form valid FOL statements.¹ Similarly, the July 2025 examination required students to assign

exact universal ($\forall$) and existential ($\exists$) quantifiers to varying statements regarding students and their affinity for books.¹ This semantic translation is treated as an absolute prerequisite competency.

Once formalization is achieved, the assessments aggressively pivot to automated inference. Chaining mechanisms—both forward and backward—are the most heavily interrogated algorithmic concepts.¹ Assessors demand not just an explanation of generalized modus ponens and the algorithms, but their successful deployment in formal proofs. Forward chaining is tested as a data-driven process, where known facts are continually applied to rules to generate new facts until a goal is met. Conversely, backward chaining is tested as a goal-driven process, working backwards from a hypothesis to find supporting axioms.³

A definitive example of this practical application is observed in the January 2025 assessment, which required candidates to formally prove the assertion "Solan is a criminal" based on a

complex matrix of textual assertions concerning hostile nations, missiles, and American citizenship.¹ Similarly, the July 2025 paper required candidates to construct a proof tree to determine that "Gita likes almond" based on a structured FOL knowledge base.¹ Underpinning these chaining mechanisms is the computational process of Unification, which is frequently targeted for explanation.¹ Candidates must understand that unification is the algorithmic procedure of finding a substitution—the Most General Unifier (MGU)—that makes different logical expressions identical, thereby permitting inference rules to fire.

Module 5: Uncertain Knowledge, Probabilistic Reasoning, and Expert Systems

The final module shifts the pedagogical paradigm from the absolute truths of deterministic logic to the nuanced realities of environments governed by stochastic elements. The integration of probability theory, specifically Bayes' Theorem, alongside the architectural implementation of Expert Systems, encapsulates the capstone competencies of the curriculum.³

Exhaustive Inventory of Historical Inquiries (Module 5)

Evaluators assess the terminal phase of the AI syllabus through complex probability computations and systems architecture design, utilizing the following historical questions.¹

Exact or Paraphrased Examination Question	Temporal Occurrence	Frequency	Bloom's Level / Course Outcome
Explain uncertain knowledge in the context of artificial intelligence. Discuss challenges with the example of diagnosing a toothache.	Jan 2025 (Q10a)	1	L2 / CO5
Explain the impact of uncertainty in probabilistic reasoning.	July 2025 (Q9a)	1	L2 / CO5
Explain basic probability	July 2024 (Q9a)	1	L3 / CO5

Notation.			
Explain Independence in Quantifying uncertainty with example.	July 2024 (Q10a)	1	L3 / CO5
Explain the inference using full joint distribution.	Jan 2026 (Q9a)	1	L2 / CO5
Explain the concept of inference using full joint probability... Calculate $P(\text{cavity} \vee \text{toothache})$, $P(\text{cavity} \text{toothache})$, $P(\sim\text{cavity} \text{toothache})$ using provided matrix.	Jan 2025 (Q10b)	1	L3 / CO5
State/Explain Baye's theorem/rule and its use/utilization in probabilistic reasoning in detail.	Jan 2025 (Q9a), Jan 2026 (Q9b), July 2024 (Q9b), July 2025 (Q9b)	4	L2, L3 / CO5
Calculate probability a person owns a dog given they walk their pet daily (based on 30% dog ownership, 70% cat, 80% walk dogs, 50% walk cats).	Jan 2025 (Q9a)	1	L3 / CO5
Write	July 2025 (Q9c)	1	L3 / CO5

representation of Bayes Theorem. Calculate fever probabilities (viral vs bacterial based on provided stats).			
Calculate the exact probability a patient has cancer given a positive test (based on base rate 0.002, 100% accuracy, 98% true positive, 97% true negative).	July 2025 (Q10b)	1	L3 / CO5
Define/Explain Expert Systems, detailing characteristics, components, capabilities, and incapacabilities. Provide examples.	Jan 2025 (Q9b), Jan 2026 (Q10a), July 2025 (Q10a)	3	L2 / CO5
Explain knowledge Acquisition in detail.	Jan 2026 (Q10b), July 2024 (Q10b), July 2025 (Q10a)	3	L2, L3 / CO5
Explain: (i) Knowledge Shell.	Jan 2026 (Q10b)	1	L2 / CO5

Theoretical Analysis and Evaluative Trends

The culmination of the artificial intelligence syllabus is dominated by rigorous quantitative probability calculations, with Bayes' Theorem functioning as the supreme evaluation mechanism.¹ Evaluators consistently deploy word problems requiring candidates to distill complex scenarios into prior probabilities, conditional likelihoods, and posterior probabilities. The emphasis is on practical calculation rather than theoretical memorization. For instance, the January 2025 examination posited a scenario wherein **30%** of a population owns dogs and

70% owns cats, introducing varying conditional probabilities for walking behaviors, and required the backward derivation of pet ownership given an observation.¹ More rigorously, the July 2025 evaluation utilized a high-stakes medical diagnostic paradigm to test Bayesian

inference: given a baseline cancer rate of $P(\text{cancer}) = 0.002$, a true positive rate of 98%, and a true negative rate of 97%, the candidate was tasked to calculate the exact probability of affliction following a positive diagnostic return using the formula

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Furthermore, the January 2025 curriculum introduced the direct, granular interrogation of the Full Joint Probability Matrix. Candidates were tasked to manually compute

$P(\text{cavity} \vee \text{toothache})$, $P(\text{cavity} \mid \text{toothache})$, and

$P(\neg \text{cavity} \mid \text{toothache})$ directly from an un-aggregated statistical table cross-referencing multiple boolean variables.¹ This calculation involves extensive marginalization and conditioning operations, proving that pure mathematical fluency is required to pass the module.

Parallel to these mathematical rigor tests is the theoretical exploration of Expert Systems.¹ Assessors consistently request descriptions detailing the interaction between the core components of these systems. Candidates must explain how domain expertise is extracted via Knowledge Acquisition, formalized into a Knowledge Base, and subsequently manipulated by an Inference Engine governed by a Knowledge Shell and exposed via a User Interface.¹

Comprehensive Topic Prognostication: The Definitive Examination Blueprint

An exhaustive examination of the data over the trailing 24-month horizon reveals several profound undercurrents governing the pedagogical architecture of the BAD402 syllabus. The cognitive demands purposefully shift from Bloom's Taxonomy Level 1 and 2 (Knowledge and Comprehension) in the early modules, to Level 3 and 4 (Application and Analysis) in the later modules. Success requires a deliberate pivot from rote memorization to algorithmic execution and mathematical logic. Based upon the exact frequency modeling and semantic analysis of the preceding test cycles, the following matrix represents the definitive blueprint of topics upon which future examination questions will be predicated.

Core I: The Fundamentals of Agency and Task Environments

The first phase of future examinations will inevitably test the boundary definitions of AI and the foundational models of agent operation. Students must prepare for theoretical questions requiring the articulation of AI milestones, the philosophy of the Turing Test, and the four main approaches to designing intelligent systems.¹

However, the primary applied metric will be the design of intelligent agents. Candidates must

be fluent in generating PEAS matrices (Performance, Environment, Actuators, Sensors) for any given real-world autonomous system, ranging from robotics to software controllers.¹ Furthermore, candidates must master the structural block diagrams of agent architectures. They must be able to visually and textually differentiate Simple Reflex Agents, Model-Based Reflex Agents, Goal-Based Agents, and Utility-Based Agents, specifically explaining how the introduction of internal state tracking resolves the limitations of partial observability.¹ Finally, the categorization of the environments themselves (Deterministic vs. Stochastic, Static vs. Dynamic, Fully vs. Partially Observable) will be tested as short-form definitional questions.¹

Core II: State-Space Modeling and Uninformed Navigation

The second core competency relies heavily on discrete mathematics and classical problem modeling. Candidates will be mandated to deconstruct abstract scenarios (specifically the 8-puzzle, Vacuum World, and Water Jug scenarios) into the five fundamental components: Initial State, Actions, Transition Model, Goal Test, and Path Cost.¹ The exam will test procedural execution; for instance, solving the Water Jug problem requires writing out exact state transitions using mathematical operators.⁹

Following state-space formulation, the examination will strictly assess uninformed search algorithms. Breadth-First Search (BFS) and Depth-First Search (DFS) are guaranteed to appear.¹ Preparation must include the ability to manually trace these algorithms through a provided graph structure, explicitly detailing the FIFO or LIFO queue states at each step. Evaluators will also demand comparative analyses regarding the time complexity, space complexity, completeness, and optimality of BFS, DFS, and Iterative Deepening DFS.¹

Core III: Heuristic Evaluation and Symbolic Logic Integration

Future assessments will utilize Module 3 to filter candidates based on algorithmic optimization. The undisputed centerpiece of this section is the A* Search Algorithm.¹ Candidates will face questions requiring them to calculate minimal pathing through a complex graph using the formula $f(n) = g(n) + h(n)$. This requires calculating the cumulative step cost and adding the provided heuristic estimates to find the optimal node expansion. Greedy Best-First

Search will also be tested, primarily as a point of contrast to prove why ignoring $g(n)$ results in suboptimal pathing.¹ Candidates must also be prepared to explain the properties of good heuristic functions, specifically the requirement for admissibility (never overestimating the cost to reach the goal).¹⁰

The secondary phase of Module 3 acts as an introduction to formal logic via the Wumpus World paradigm.¹ Candidates must understand how to represent percepts logically and navigate the environment. This necessitates a fundamental understanding of Propositional Logic, including logical equivalence rules, atomic versus complex sentences, Horn clauses, and the deployment of basic connectives.¹

Core IV: First-Order Logic and Algorithmic Inference

The fourth competency area strictly assesses the candidate's fluency in predicate logic. Examinations will require the semantic translation of complex English sentences into

First-Order Logic (FOL) utilizing Universal ($\forall$) and Existential ($\exists$) quantifiers, predicates, and constants.¹ Mastery over classic deductive scenarios (such as the "Marcus was a Pompeian" syntax structure) is non-negotiable.¹

Once the knowledge base is formalized, the exam will test the automated reasoning mechanisms. Unification—the process of standardizing variables to permit logical resolution—will be heavily targeted.¹ The capstone of this module is the application of Forward Chaining and Backward Chaining.¹ Candidates must be prepared to execute a formal proof. They will be given a textual scenario, required to map it into FOL axioms, and then expected to build a deductive chain (either data-driven or goal-driven) to logically prove a final assertion.

Core V: Probabilistic Architecture and Expert Systems

The final blueprint component transitions entirely to stochastic mathematics. Bayes' Theorem is the supreme evaluation vector.¹ Future exams will invariably present diagnostic word problems (e.g., medical testing, fault detection) requiring candidates to extract the prior probability, true positive rate, and false positive rate, and plug them into the Bayesian equation to derive the posterior probability.¹

Furthermore, candidates must understand Full Joint Probability Distributions and be capable of executing marginalization and conditional probability calculations across a raw matrix of boolean variables.¹ To conclude the assessment cycle, the operational architecture of Expert Systems will be tested.¹ Candidates must be able to diagram the flow of information from a human expert, through Knowledge Acquisition mechanisms, into a structured Knowledge Base, which is then manipulated by an Inference Engine within a Knowledge Shell to provide probabilistic advice to end-users.¹

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